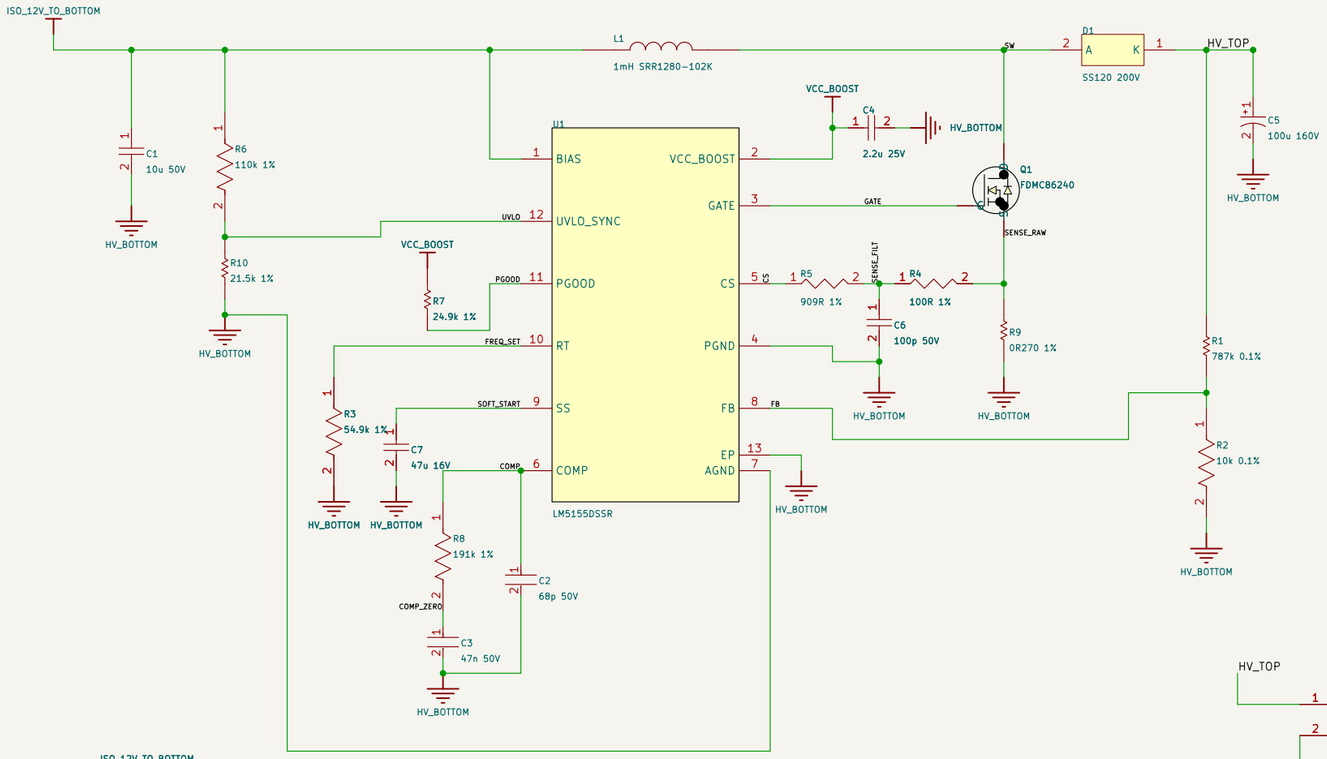
LM5155 BOOST — ELECTRICAL DRAFT

DRAFT — BENCH PROTOTYPE. 12 V isolated input / 79.7 V nominal output.

Conditional bench target 15 mA. GAPTEC + 200 ohm preload cannot be assumed to support that load.

HV_BOTTOM = GAPTEC pin 5 = rail-splitter negative input. It is NOT signal ISO_GND.



HIGH VOLTAGE RETURN MAP
J1.2 and J2.2 = HV_BOTTOM.
J2.1 = HV_TOP.
Signal ISO_GND connects only to
rail-splitter MID downstream.

PROVISIONAL LOOP NETWORK
R8 = 191k; C3 = 47n; C2 = 68p.
Simplified 500 Hz crossover target.
Actual phase margin, source/load
interaction and ESR are unvalidated.

SOFT START
C7 = 47u nominal; require effective
C >= 33u over 0-1 V. About 4 s nominal
boost rise with 100u output bulk.
Startup load must be controlled.

SAFETY / INTERFACE LIMITS
Stopping switching does not disconnect
input from output. 100u at 80 V stores 0.32 J.
PGOOD pullup is local VCC (~6.85 V),
not a Pico or MID-referenced signal.
No fault-safe output disconnect here.

R1/R2 set 79.7 V. R9 = 0.270 ohm current sense replaces conflicting source metadata.
L1 = SRR1280-102K, 1 mH. Q1 = FDMC86240, 150 V, specified at 6 V gate drive.
Exact R/C MPNs selected; effective capacitance and loop behavior require bench acceptance.

Before powered integration: loop model/bench steps, inrush/UVLO, efficiency and
GAPTEC 2 W budget, switch-node peak stress/snubber, thermal behavior, discharge/protection, isolation layout.

Bench prototype — compensation and integrated loading require validation

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Title: 12 V to 79.7 V boost — engineering bench prototype

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